

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 16

Pharynx & Larynx

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Recommended reading

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 5th ed. Elsevier. Chapter 8.



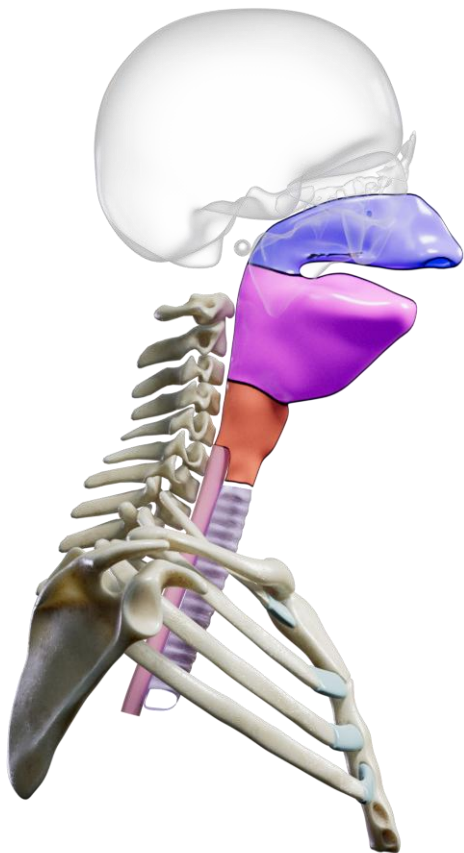
Pay special attention to the 'In the Clinic' boxes which contain the clinical correlates for this topic.



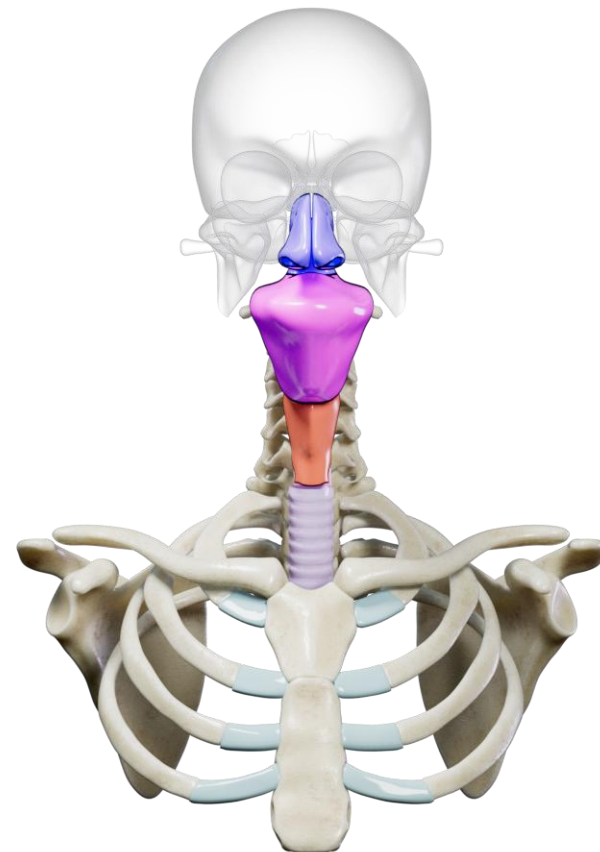
Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0124	Delineate the borders between the naso-, oro-, and laryngo-pharynx
SOM.MKII.BPM2.1.NB.1.ANAT.0125	List the general attachments, innervations and functions of the muscles of the pharynx
SOM.MKII.BPM2.1.NB.1.ANAT.0126	List the contributions to the pharyngeal nerve plexus
SOM.MKII.BPM2.1.NB.1.ANAT.0064	Describe the fascial layers of the neck including investing fascia, pretracheal and buccopharyngeal fascia, prevertebral fascia and alar fascia and their clinical relevance
SOM.MKII.BPM2.1.NB.1.ANAT.0128	Explain the sensory and motor innervation of the nasopharynx, oropharynx, and laryngopharynx
SOM.MKII.BPM2.1.NB.1.ANAT.0129	Describe the piriform fossa, its innervations and clinical importance
SOM.MKII.BPM2.1.NB.1.ANAT.0130	Describe the arterial supply of the pharynx, and its venous and lymphatic drainage
SOM.MKII.BPM2.1.NB.1.ANAT.0131	Describe the location of the pharyngeal, palatine, and lingual tonsils and the clinical importance of this tonsillar arrangement
SOM.MKII.BPM2.1.NB.1.ANAT.0132	Identify the hyoid bone and the cartilages that form the larynx
SOM.MKII.BPM2.1.NB.1.ANAT.0133	Identify the thyrohyoid and cricothyroid membranes
SOM.MKII.BPM2.1.NB.1.ANAT.0134	Define the boundaries and extent of the larynx
SOM.MKII.BPM2.1.NB.1.ANAT.0135	Describe the true and false vocal cords, ventricle, vestibule, and rima glottidis
SOM.MKII.BPM2.1.NB.1.ANAT.0136	List the attachments, innervation and functions of the muscles of the larynx
SOM.MKII.BPM2.1.NB.1.ANAT.0137	Describe the sensory and mucus innervation of the larynx
SOM.MKII.BPM2.1.NB.1.ANAT.0138	Describe the arterial supply of the larynx, and its venous and lymphatic drainage
SOM.MKII.BPM2.1.NB.1.ANAT.0139	Compare and contrast the clinical picture of a patient with a loss of recurrent versus internal versus external laryngeal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0255	Describe the clinical importance of the arrangement of pharyngeal, palatine and lingual tonsils
SOM.MKII.BPM2.1.NB.1.ANAT.0141	Explain how the pharynx and larynx function during deglutition
SOM.MKII.BPM2.1.NB.1.ANAT.1168	Compare and contrast structures incised during a tracheotomy versus a cricothyrotomy procedure
SOM.MKII.BPM2.1.NB.1.ANAT.0230	Identify major anatomical structures of the pharynx in MRI, CT and X-ray imaging

Overview



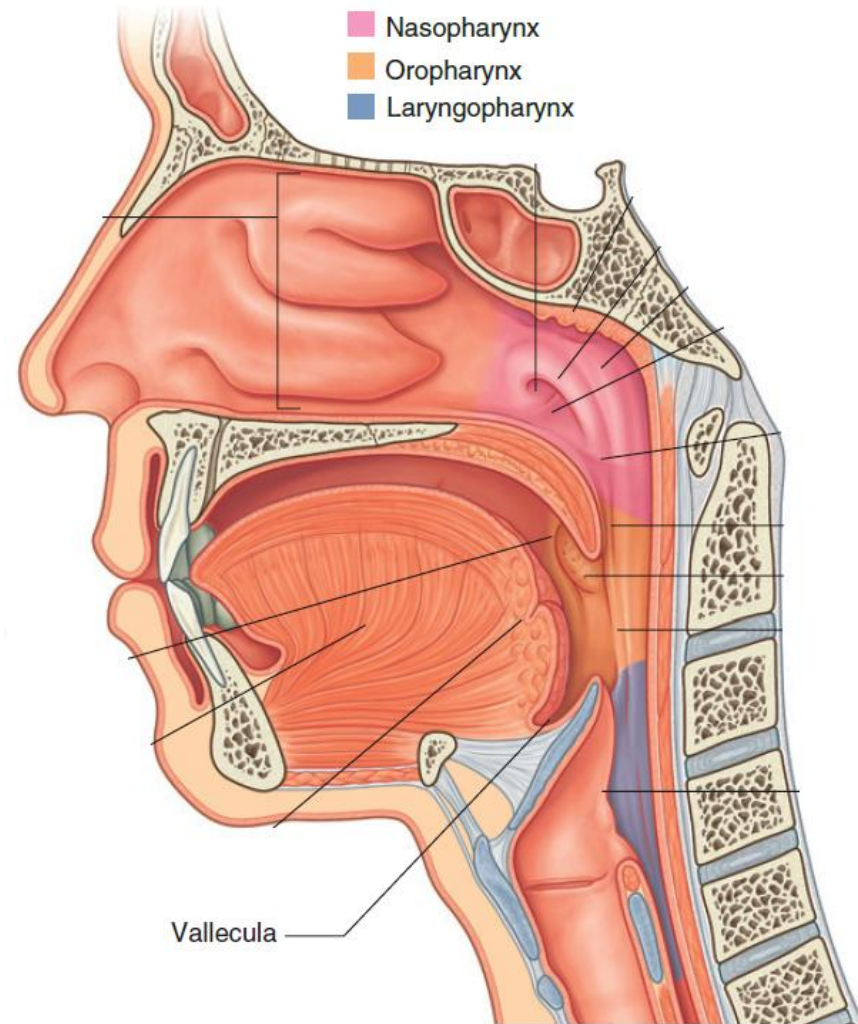
- Nasopharynx,
Oropharynx
- Laryngopharynx
- Larynx
- *Cranial nerves:*
V, IX, X



-  Nasal Cavity/ Nasopharynx
-  Oral Cavity/ Oropharynx
-  Laryngopharynx

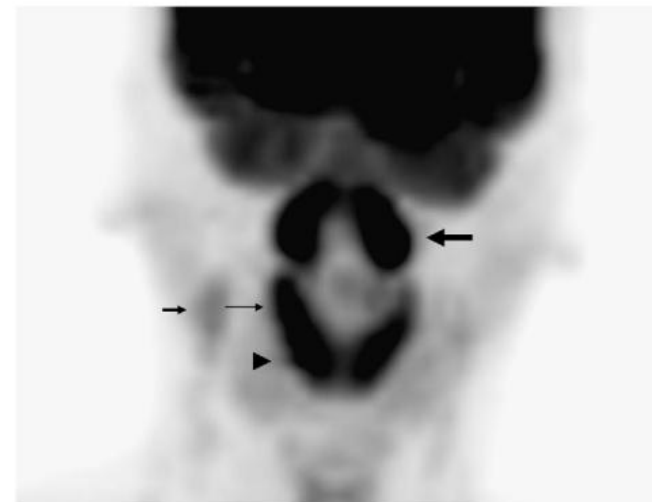
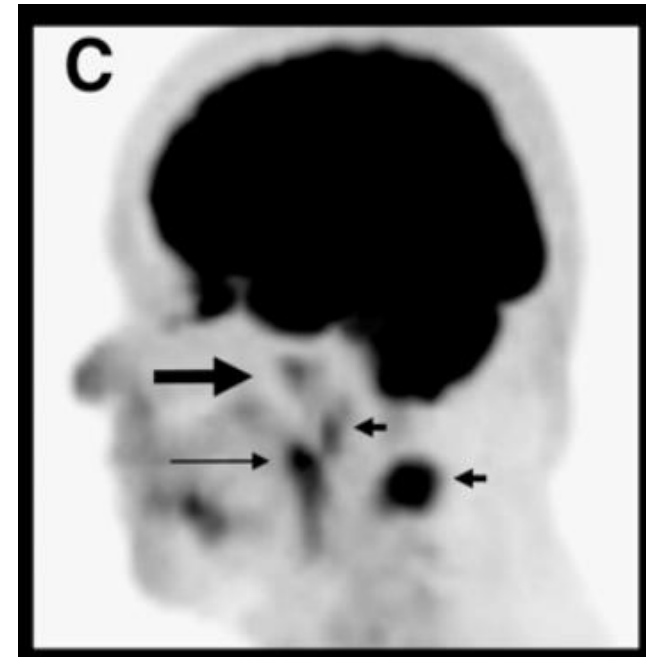
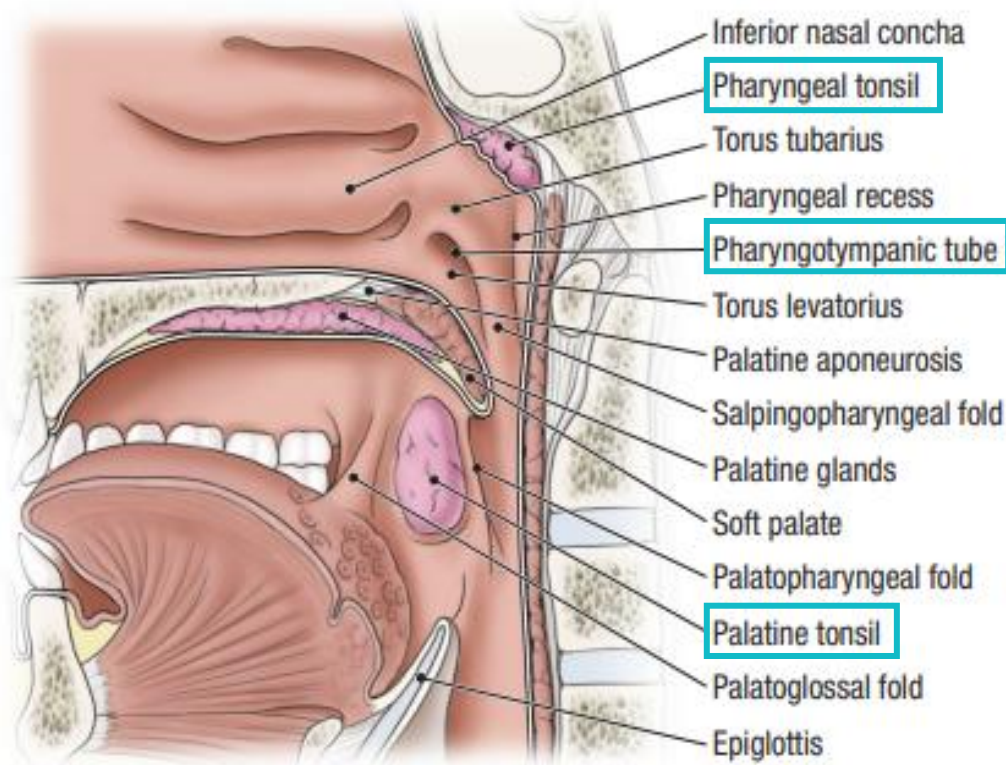
Pharynx

- Superior expansion of alimentary system
 - extends from cranial base to C6 vertebral level
- Links nasal and oral cavities with esophagus and larynx
 - **Nasopharynx**: between sphenoid bone and soft palate
 - **Oropharynx**: between soft palate and base of tongue
 - **Laryngopharynx**: between epiglottis and C6
- Serves as a common pathway for air and food

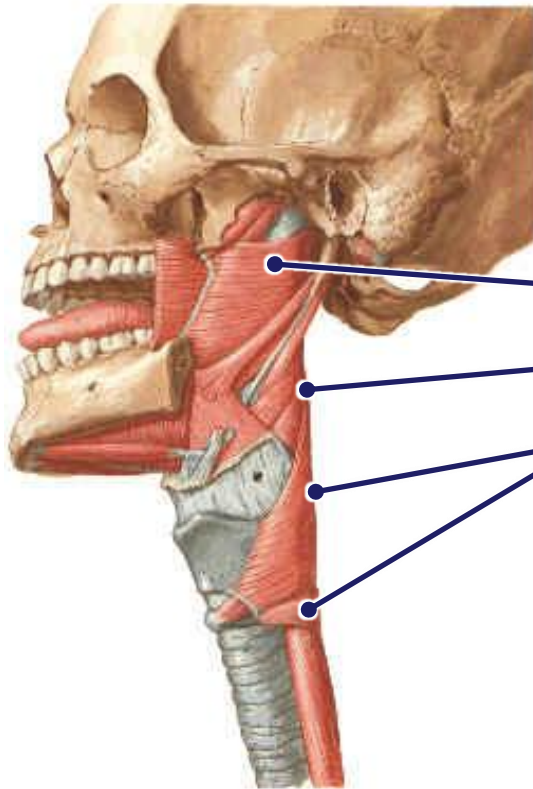


Tonsillar Ring

- Lymphoid collections in the superior pharynx
 - *Pharyngeal tonsil* – in posterior nasopharynx
 - called "adenoids" when enlarged
 - *Tubal tonsils* – near pharyngeal orifice of pharyngotympanic tube
 - *Palatine tonsils* – between faucial arches
 - *Lingual tonsil* – in root of the posterior tongue



Pharynx - Circular Muscles



- Joined in the midline at **pharyngeal raphae**

3 groups:

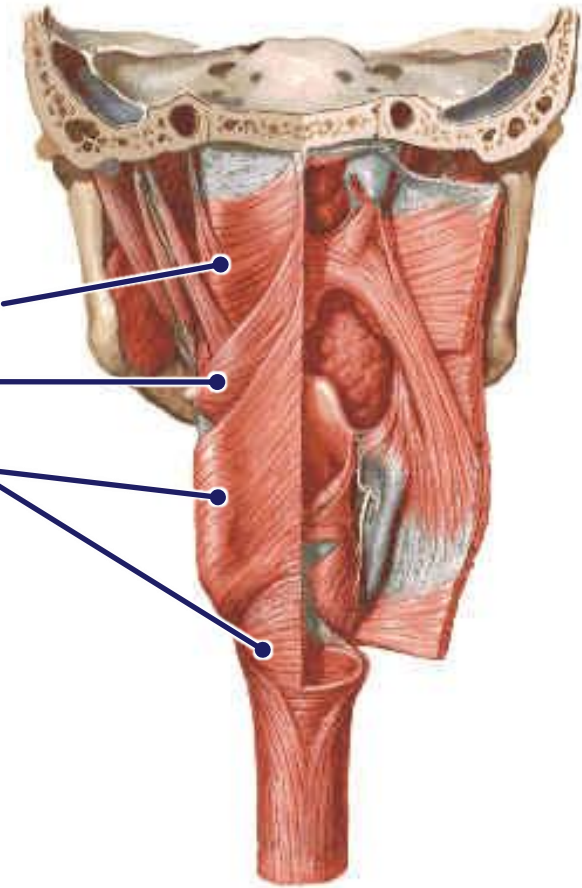
superior constrictors

middle constrictors

inferior constrictors

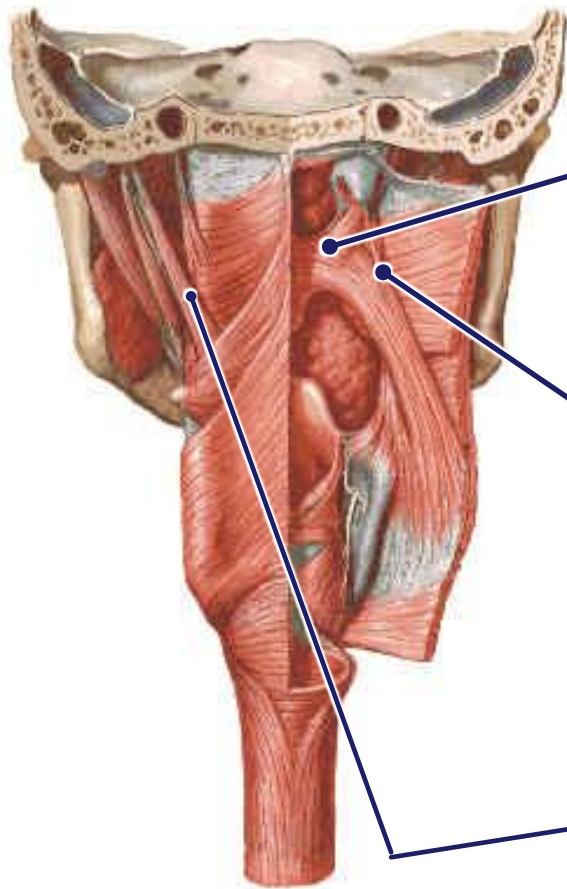
- Function:
constrict the pharyngeal lumen to propel a bolus downward during swallowing

- Innervated by **X**



Pharynx - Longitudinal Muscles

- Function:
elevate and shorten
the pharynx during
swallowing



Palatopharyngeus

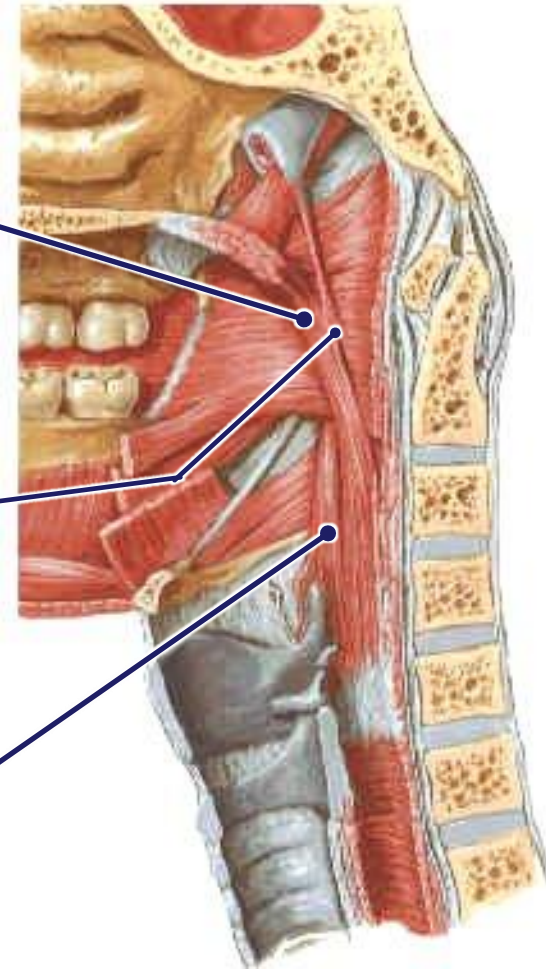
- supplied
by **pharyngeal
branch of X**

Salpingopharyngeus

- pulls on *auditory
tube*
- supplied by
**pharyngeal
branch of X**

Stylopharyngeus

- supplied by **IX**

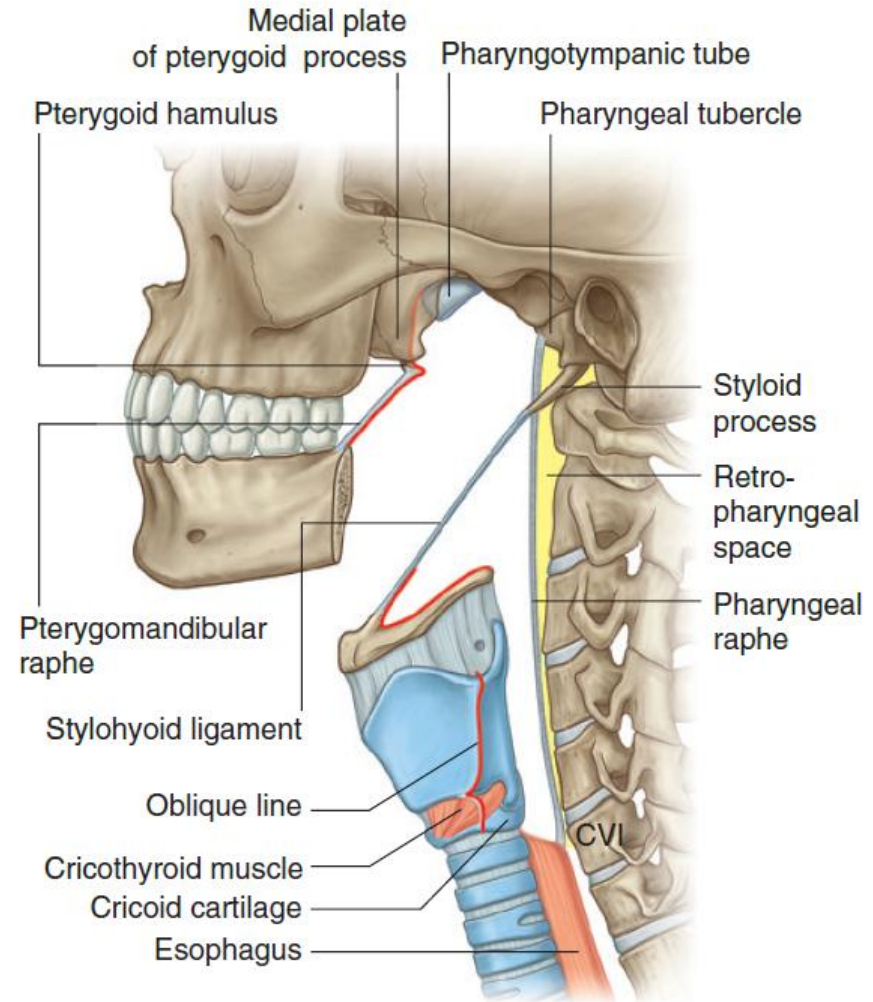


Lateral Pharyngeal Wall

- Lateral attachment of pharyngeal wall is discontinuous

Three parts:

- Medial pterygoid plate to pterygomandibular raphe
- Lower stylohyoid ligament to greater horn of hyoid bone
- Oblique line from thyroid cartilage to cricoid cartilage

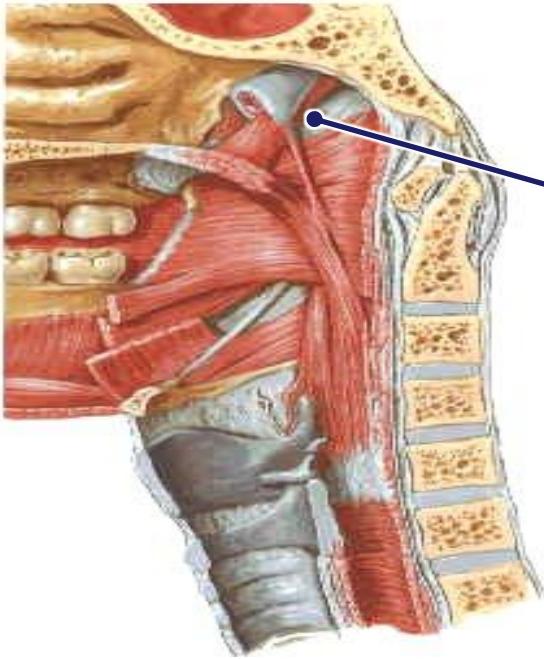


Pharyngeal Fascia

- Surrounds pharyngeal muscles
- Reinforces gaps in pharyngeal wall

Two layers

- Pharyngobasilar fascia: inner aspect
- Buccopharyngeal fascia: outer aspect



Oropharynx —————

Buccopharyngeal fascia —————

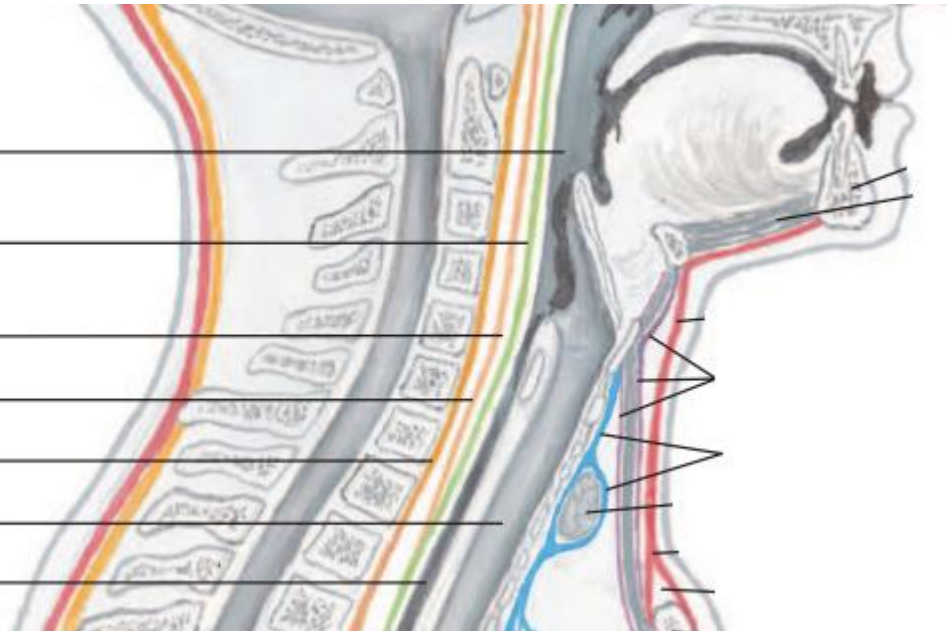
Retropharyngeal space —————

Alar fascia —————

Prevertebral fascia —————

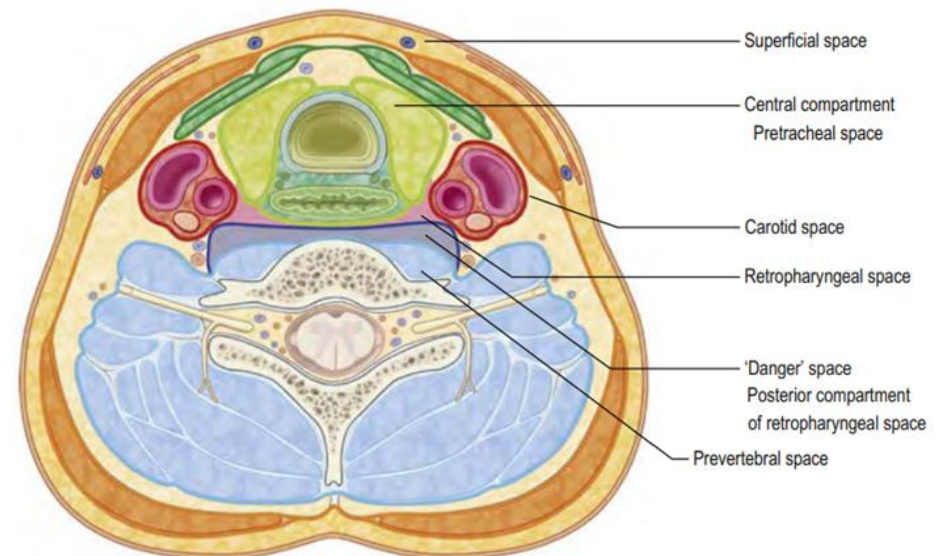
Trachea —————

Esophagus —————



Retropharyngeal Space

- Between
 - Buccopharyngeal fascia &
 - Visceral prevertebral layer of deep cervical fascia
- Largest interfascial space in the neck
 - loose CT that can act as a potential space
 - Possible passageway for the spread of infections or masses into the mediastinum
- Can lead to
 - difficulty swallowing (**dysphagia**)
 - difficulty speaking (**dysarthria**)

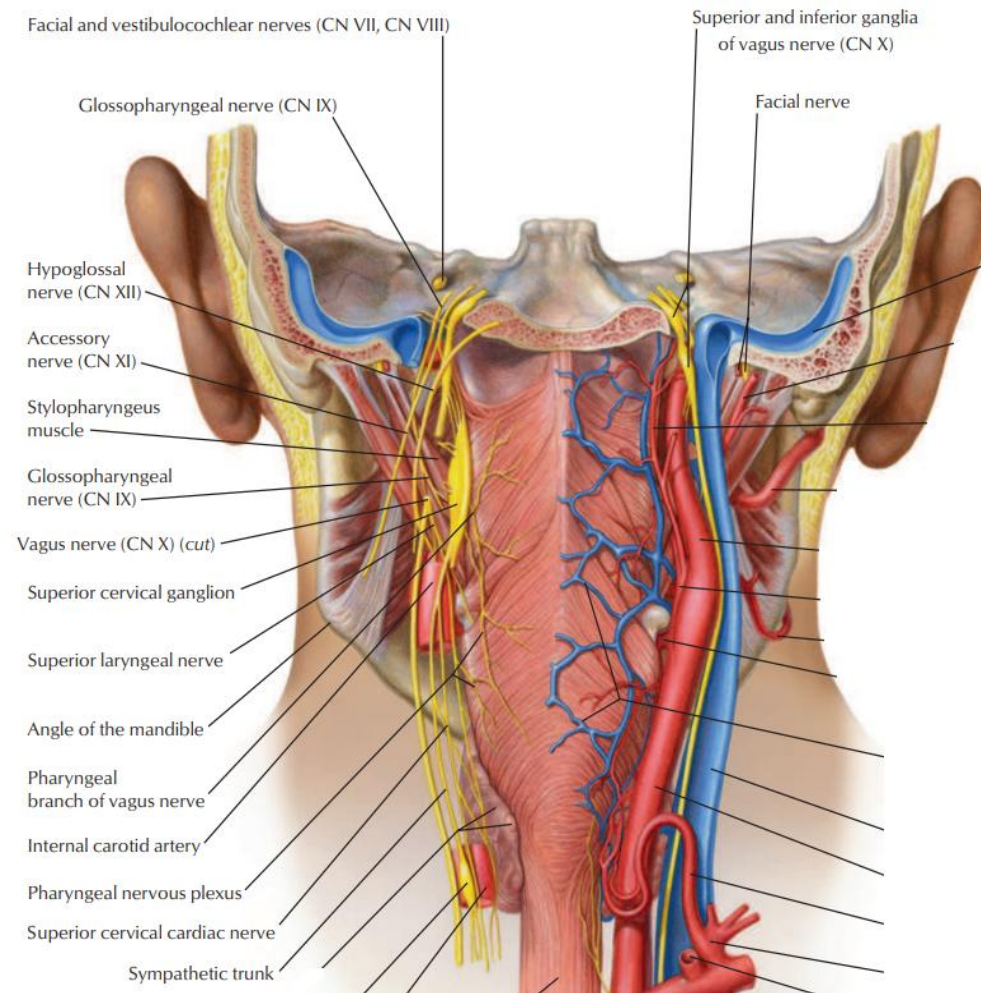


Pharyngeal Nerve Plexus

- Derived from
 - pharyngeal branch of IX
 - pharyngeal branch of X
 - superior cervical sympathetic ganglion

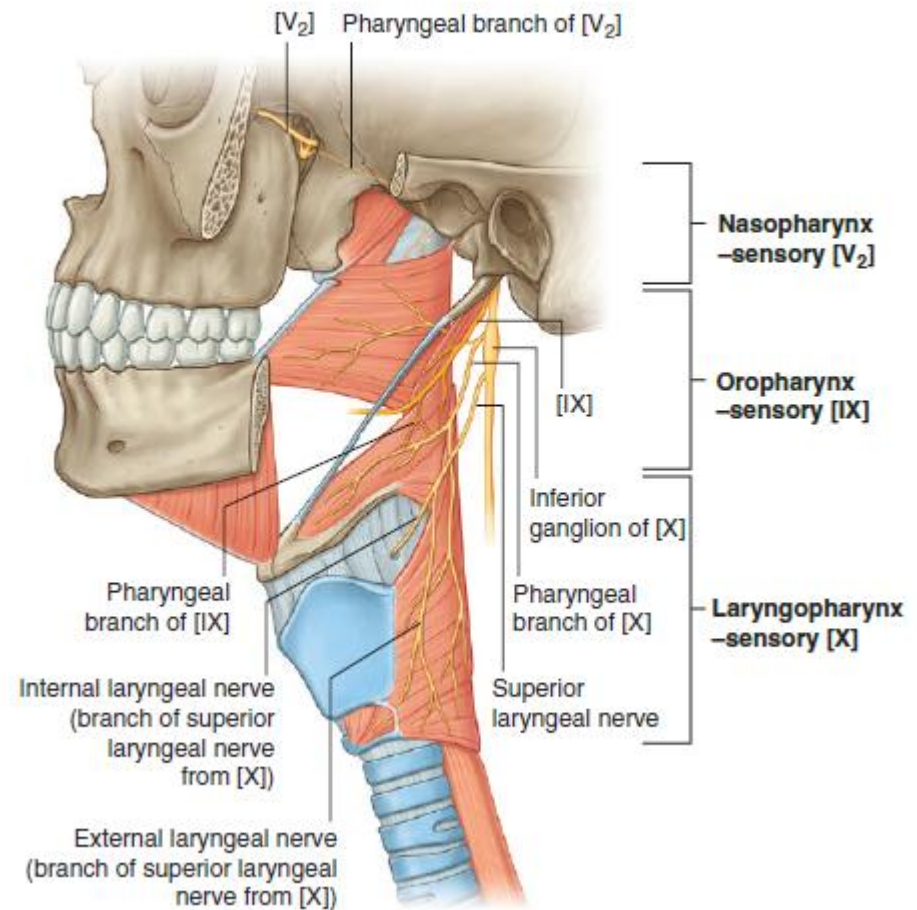
General guide:

- Provides **motor and sensory** innervation to most of the pharynx
 - excluding *nasopharynx*



Pharynx - Sensory Innervation

- *Nasopharynx*
 - mostly pharyngeal branch of **V2**
- *Oropharynx*
 - mostly **IX** via **pharyngeal plexus**
 - afferent limb of **gag reflex**
- *Laryngopharynx*
 - mostly **X** via **internal laryngeal nerve**



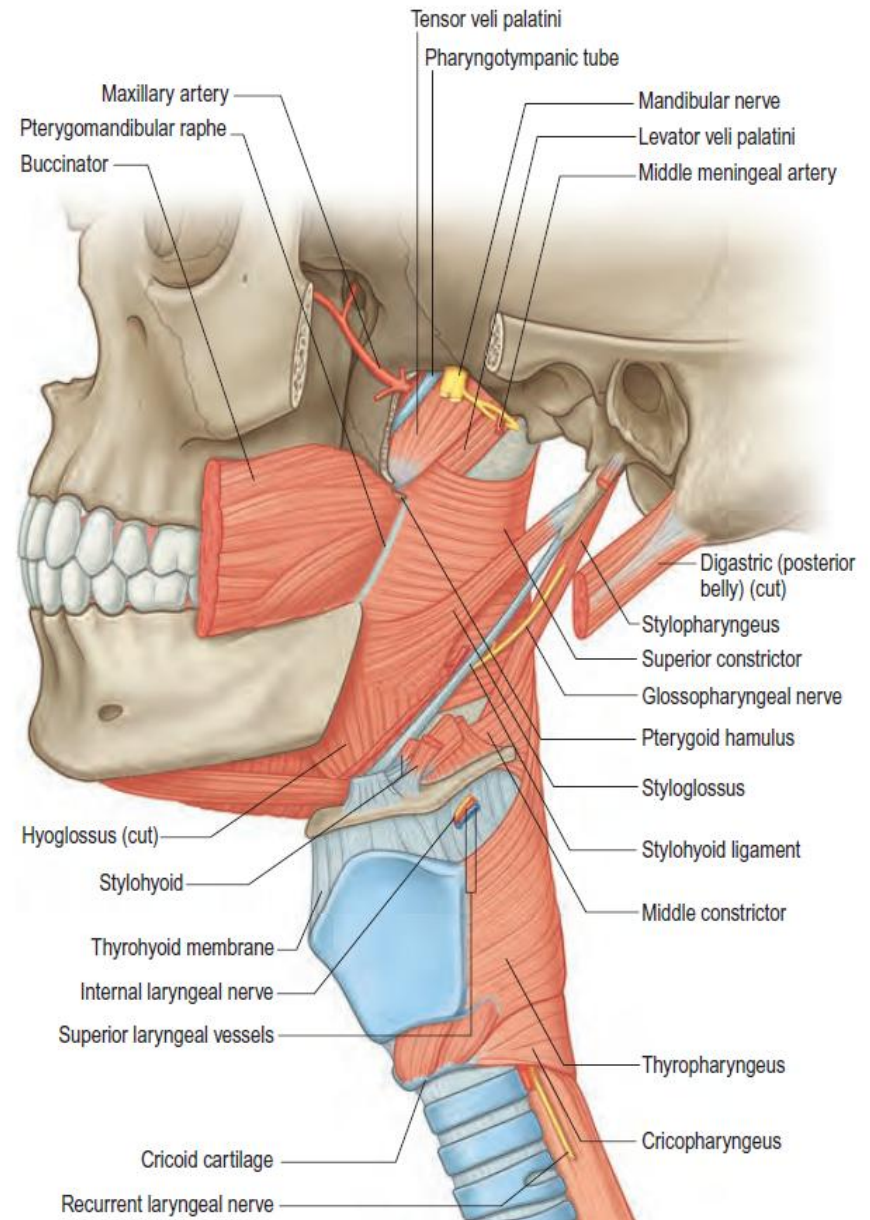
Pharynx – Motor Innervation

Mostly **pharyngeal branch of X**

- efferent limb of **gag reflex**
 - e.g. glottis closure, elevated soft palate

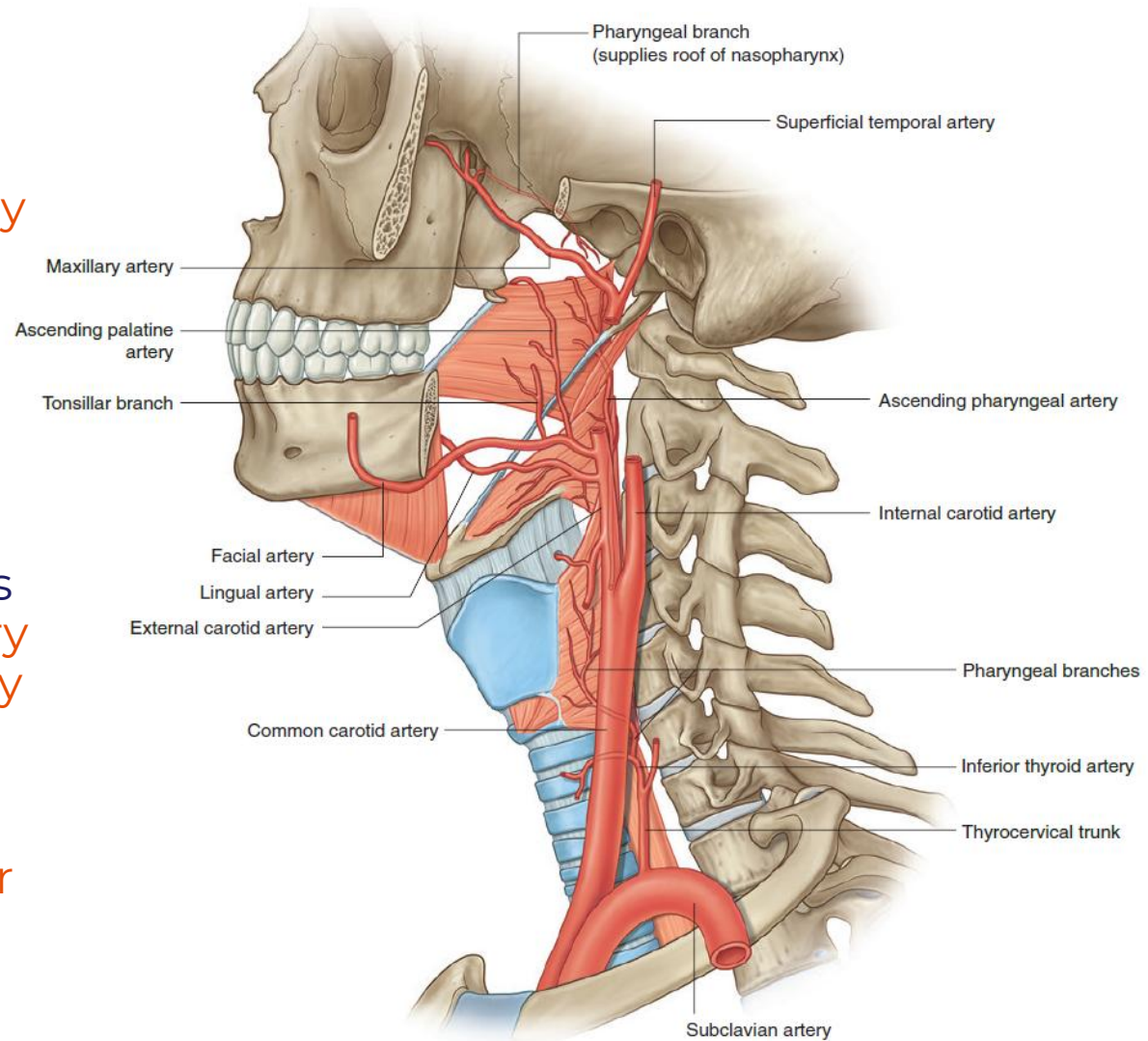
Exception:

- **Stylopharyngeus**
 - innervated by **IX**



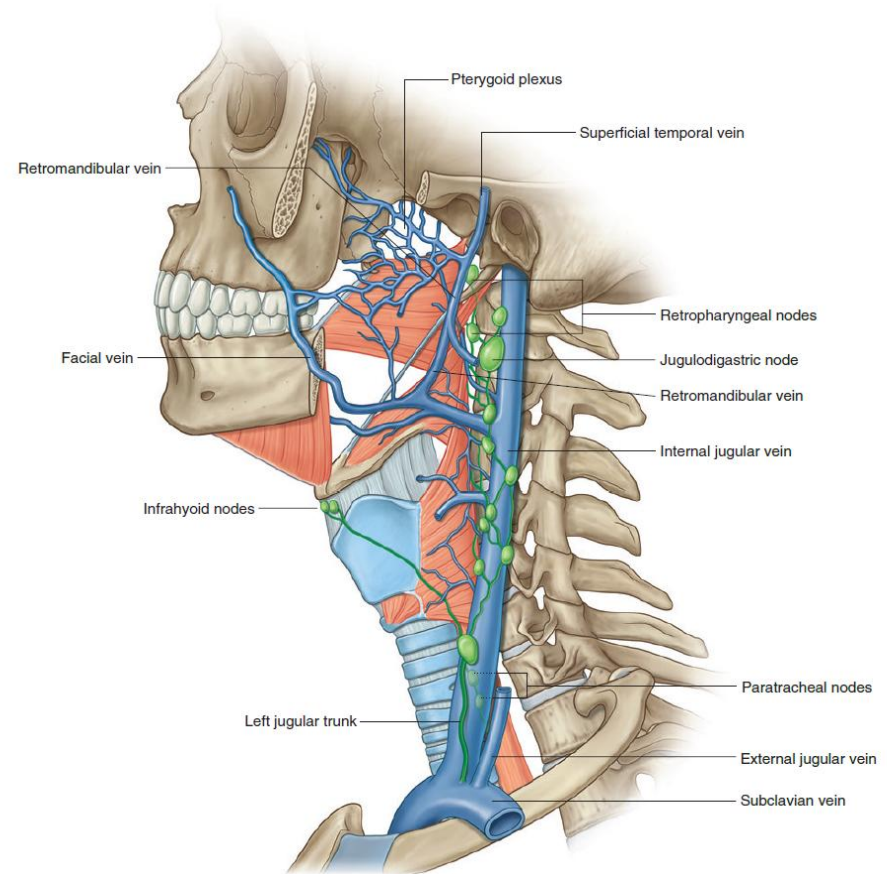
Blood supply of the Pharynx

- External carotid artery
 - Ascending pharyngeal artery
 - Ascending palatine artery and tonsillar branches of facial artery
 - Numerous smaller branches of maxillary artery and lingual artery
- Subclavian artery
 - Pharyngeal branch of inferior thyroid artery



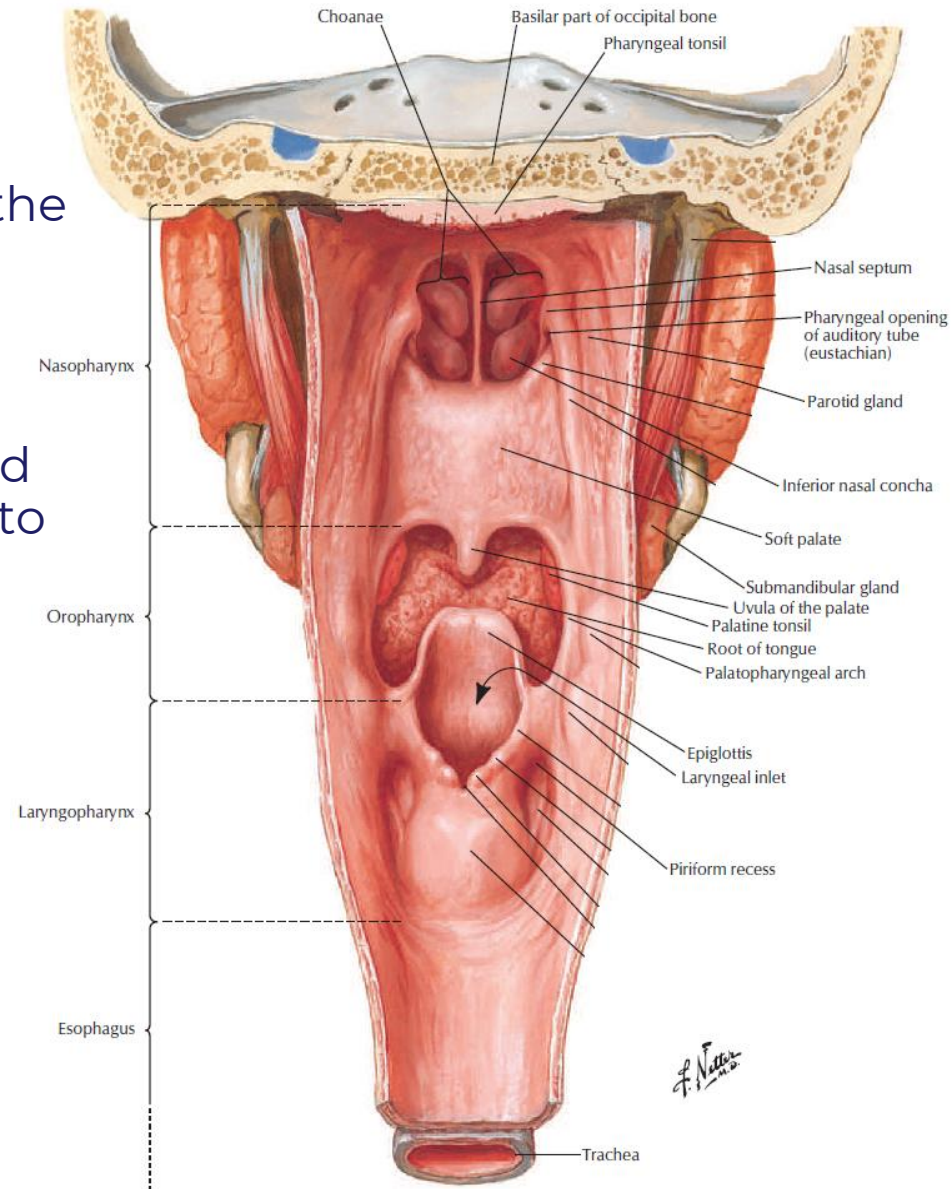
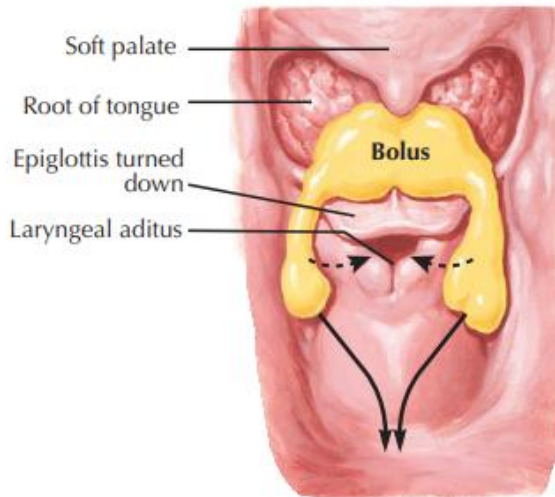
Drainage of the Pharynx

- Venous drainage
 - Veins of the pharynx form a plexus
 - Superiorly drains into **pterygoid plexus**
 - Inferiorly drains into **facial vein** and **internal jugular vein**
- Lymphatic drainage
 - Enters **deep cervical nodes**, including
 - *Retropharyngeal*
 - *Paratracheal*
 - *Infrathyoid*
 - Palatine tonsils drain into **jugulodigastric nodes**



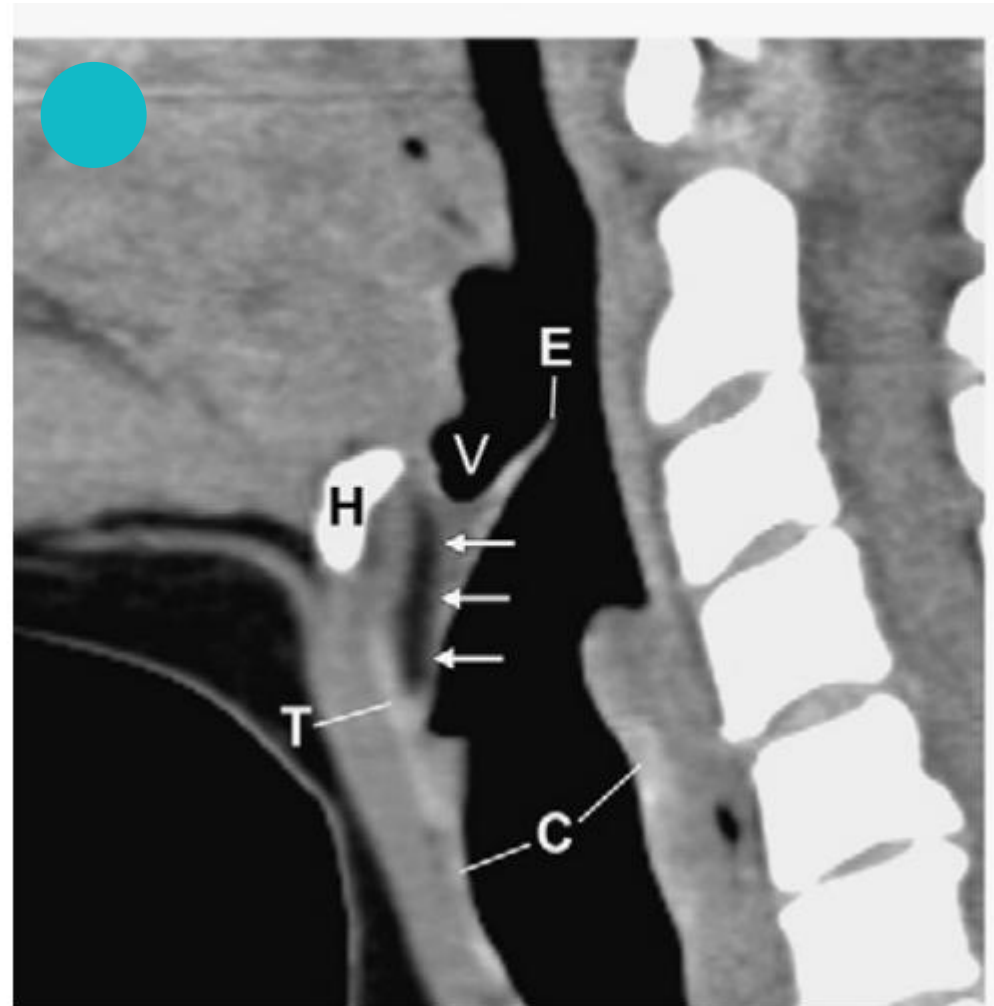
Laryngopharynx

- From superior margin of epiglottis to top of esophagus
- Anterior wall has opening into the laryngeal inlet
- contains **Piriform fossae**
 - mucosal recesses
 - direct ingested food around the raised laryngeal inlet into the esophagus
 - AKA "*smuggler's fossa*"



Larynx

- Musculoligamentous structure with a cartilaginous framework
- Extends from tongue to trachea
 - function as an *air passage*
- Forms anterior wall of laryngopharynx
 - function as a *sphincter*
- Covered by skin, fasciae and infrahyoid strap muscles
 - function as an *organ of phonation*

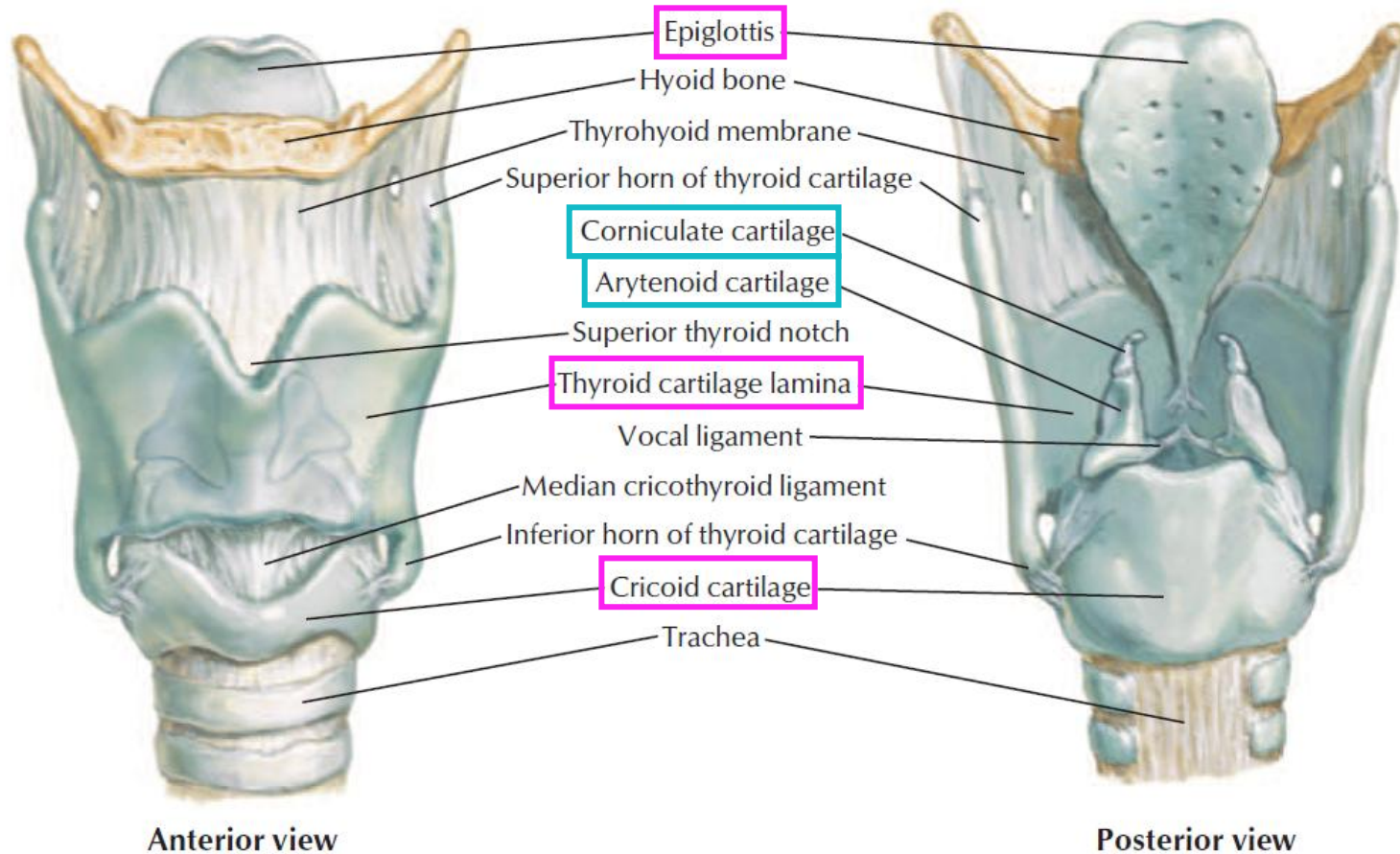


doi: 10.1016/j.otc.2012.08.006

C: cricoid
T: thyroid cartilage
H: hyoid bone
E: epiglottis
V: vallecula

Larynx - Cartilages

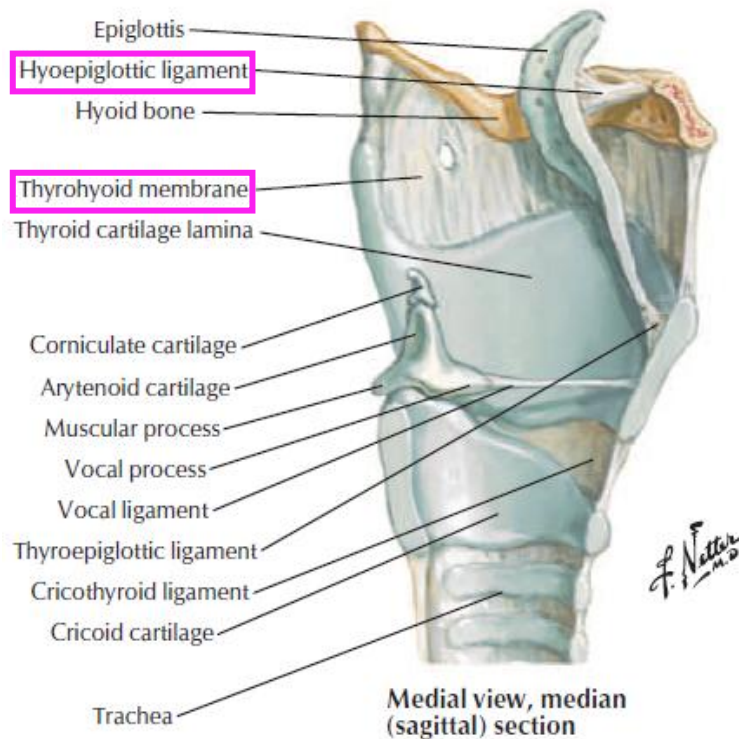
- 3 large unpaired cartilages:
cricoid, thyroid (with laryngeal prominence), epiglottis
- 3 smaller pairs of cartilage:
arytenoid, corniculate, cuneiform (in quadrangular membrane)



Larynx - Ligaments

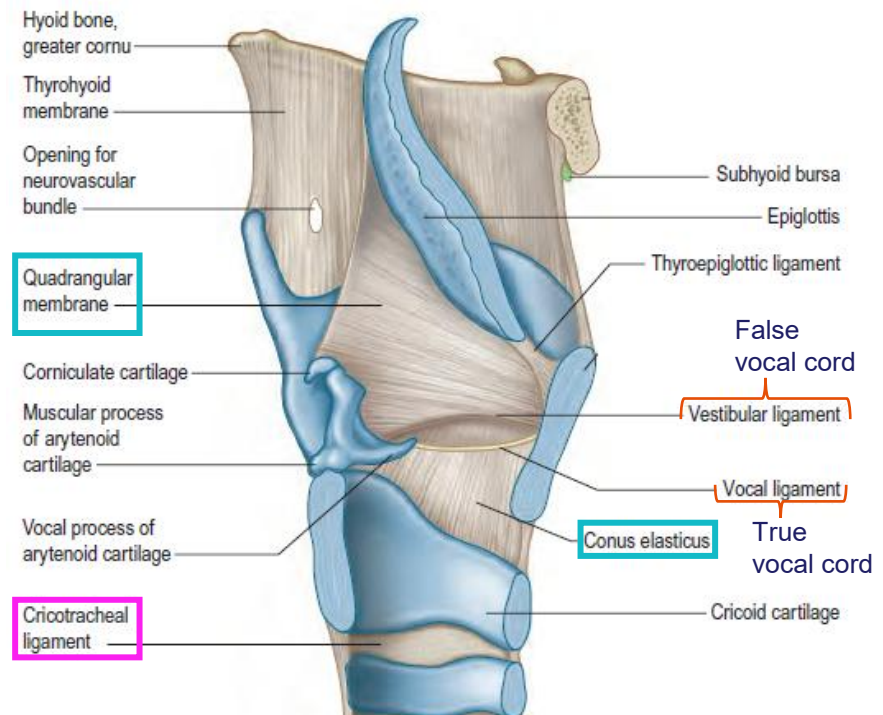
Extrinsic ligaments:

- Hyo-epiglottic ligament
- Thyrohyoid membrane
- Cricotracheal ligament



Intrinsic ligaments: fibroelastic membrane with 2 parts

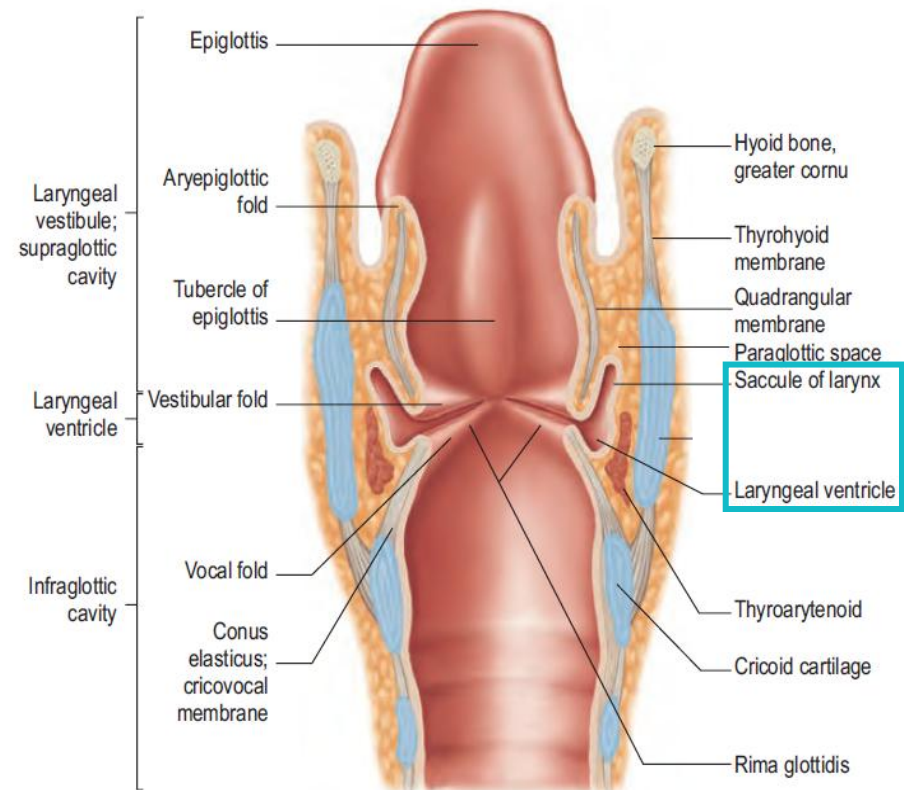
- Upper – Quadrangular membrane
- Lower – Conus elasticus



Cavity of the Larynx

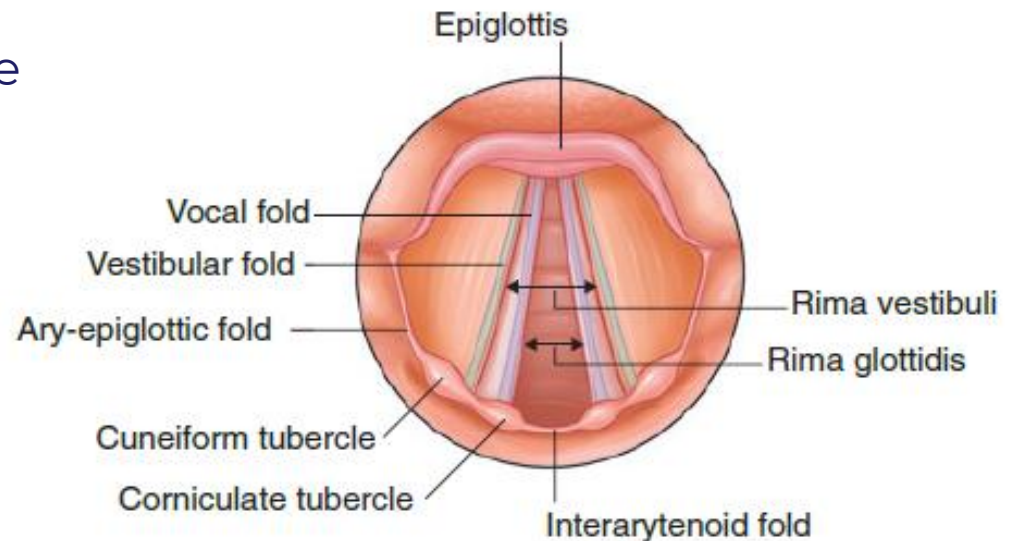
Vestibular and Vocal folds constrict and divide the central cavity into 3 regions:

- **Vestibule**
- **Middle chamber:**
between vestibular and vocal folds
 - *Ventricle*: mucosal bulges between folds
 - *Saccules*: tubular extension of ventricles
- **Infraglottic cavity**

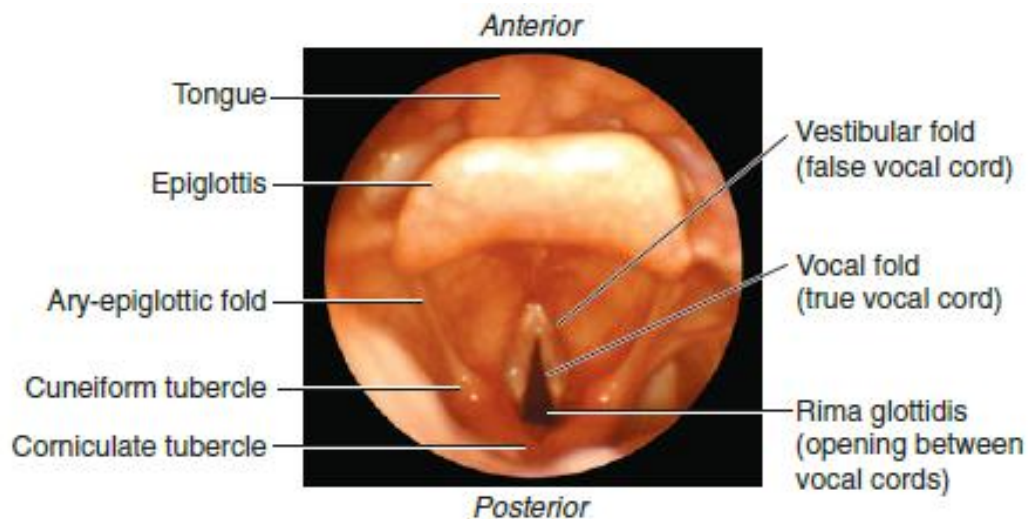
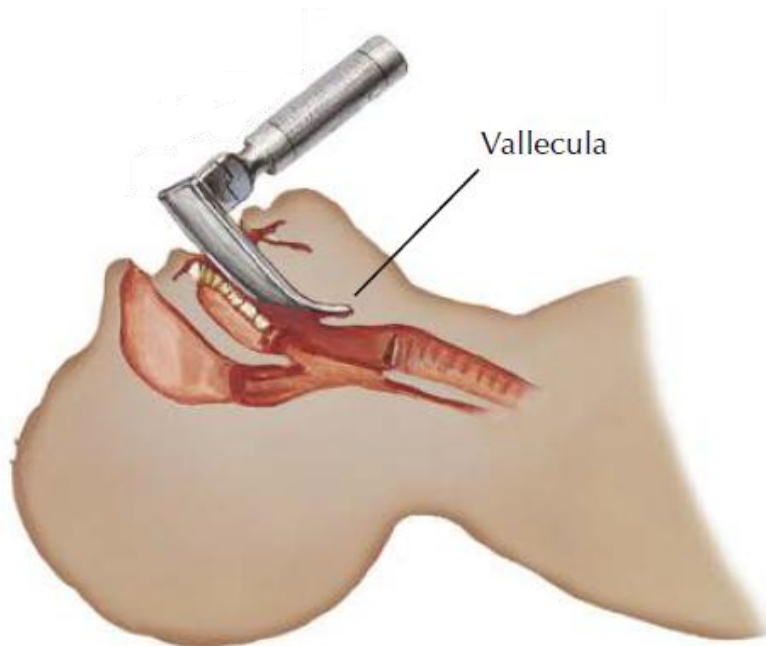


Laryngeal Cavity – Viewed from Above

- **Rima vestibuli:**
triangular opening
between two adjacent
vestibular folds
 - at the entrance to the
middle chamber
- **Rima glottidis:**
narrower triangular
opening between the
vocal folds.
 - base formed by
interarytenoid fold

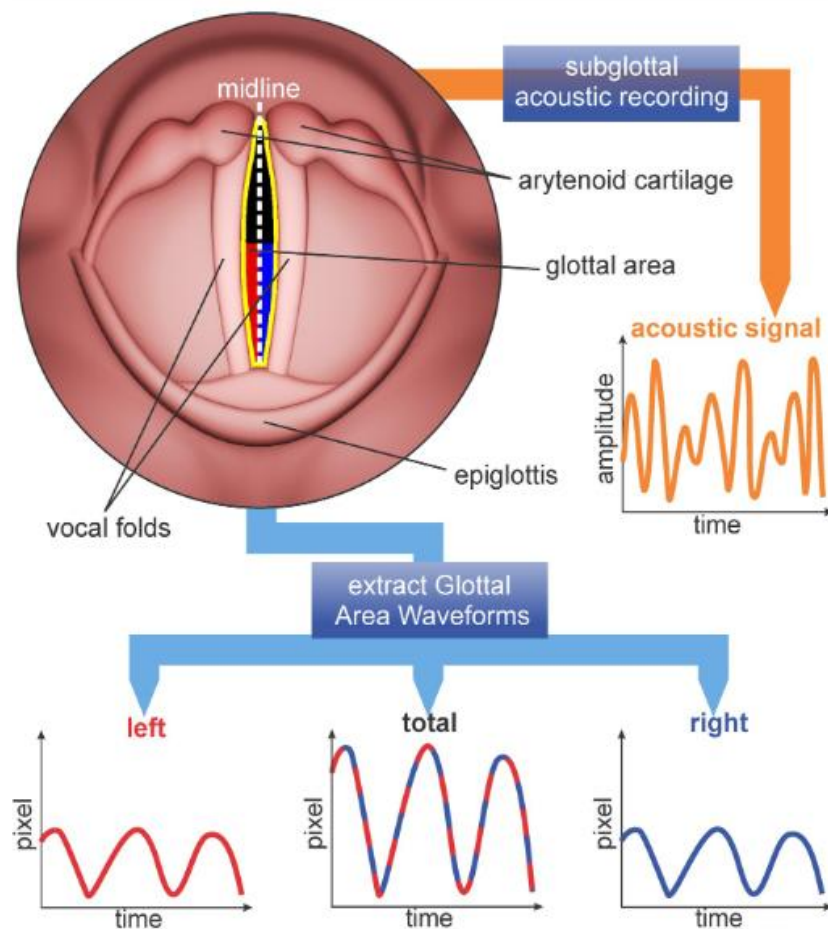


Direct Laryngoscopy View



Speech Production

- Muscles position and tension vocal folds
- Subglottal vibrations produce sound
- Sounds are recognized as speech



Intrinsic Muscles of the Larynx

Muscles:

- Cricothyroid
- Posterior and Lateral crico-arytenoid
- Transverse and Oblique arytenoid
- Thyro-arytenoid
- Vocalis

Innervation:

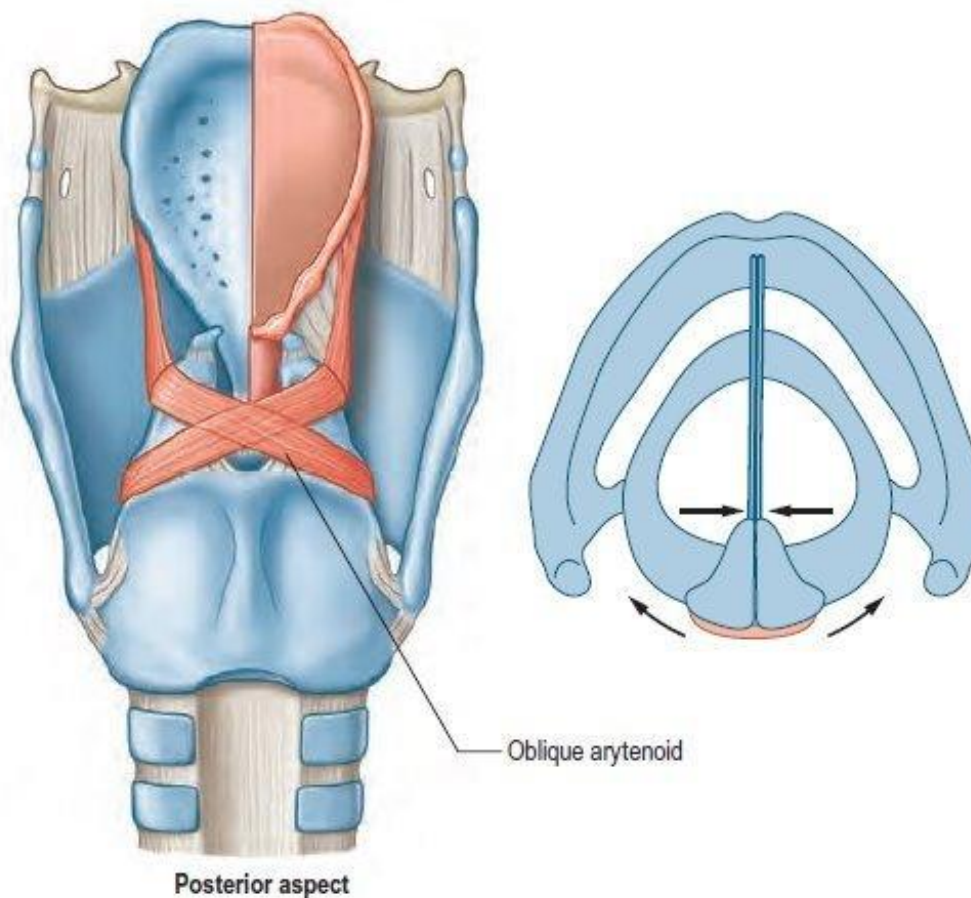
- Cricothyroid:
external laryngeal nerve (X)
- All others (interarytenoid):
recurrent laryngeal (X)

Actions:

- Pull on vocal ligaments
- Open/close rima glottidis
- Control dimensions of the vestibule
- Close rima vestibuli
- Close laryngeal inlet

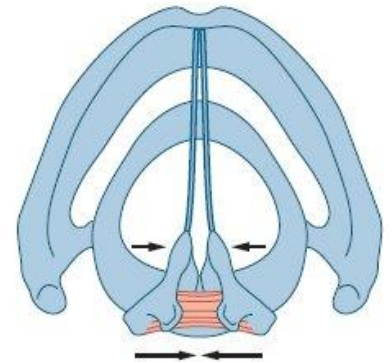
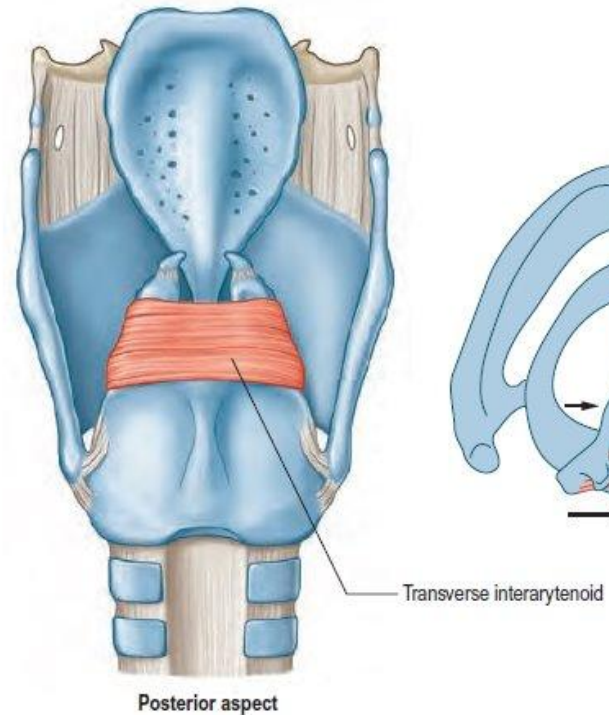
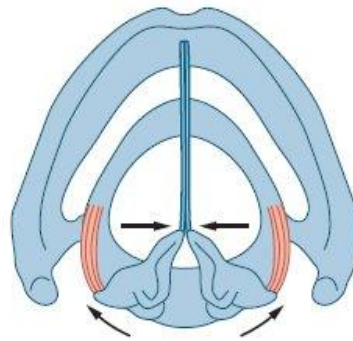
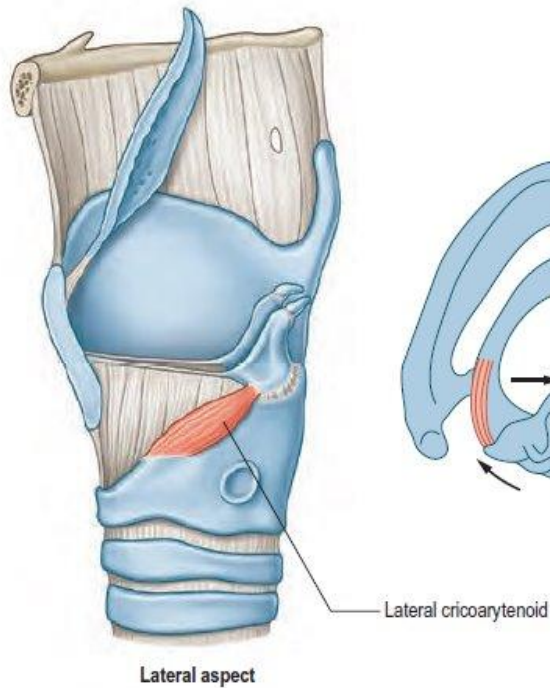
Oblique arytenoid muscles

- Approximate arytenoid cartilages to the midline of the epiglottis
 - acts as sphincter of the laryngeal inlet



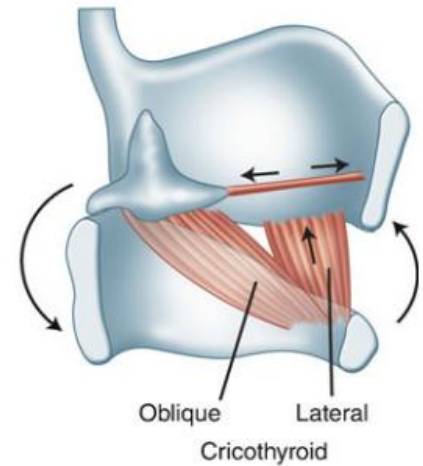
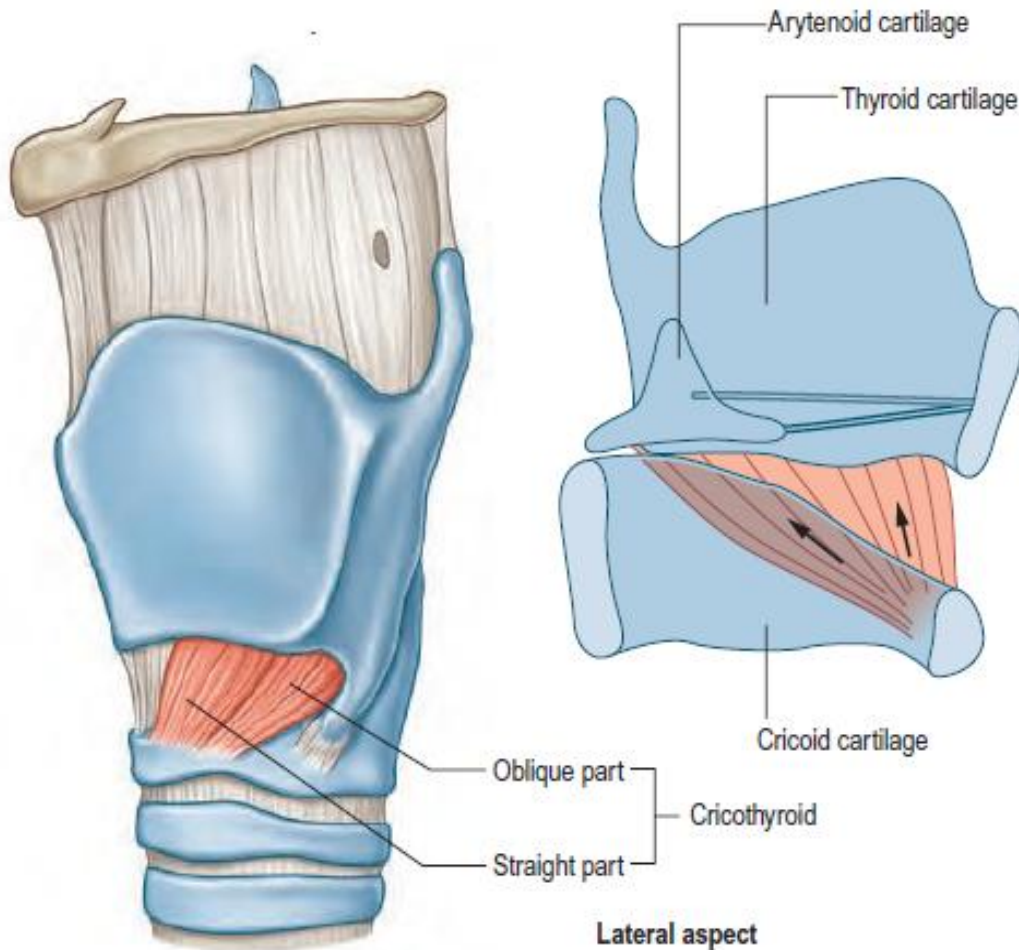
Lateral cricoarytenoid & Transverse arytenoid muscles

- Adduct arytenoid cartilages
 - close rima glottidis



Cricothyroid muscles

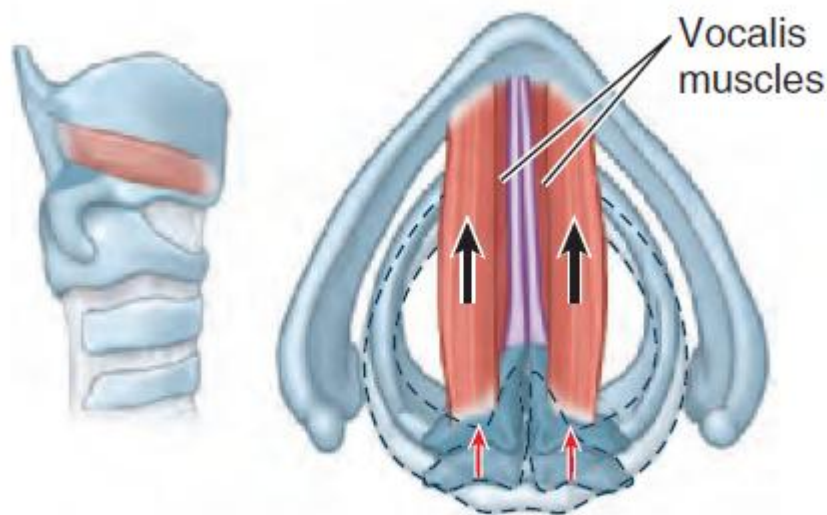
- Lengthen vocal folds
 - increases vocal fold tension (increases vocal pitch)



DOI: 10.1007/978-1-4614-3794-9_13

Thyro-arytenoid & Vocalis muscles

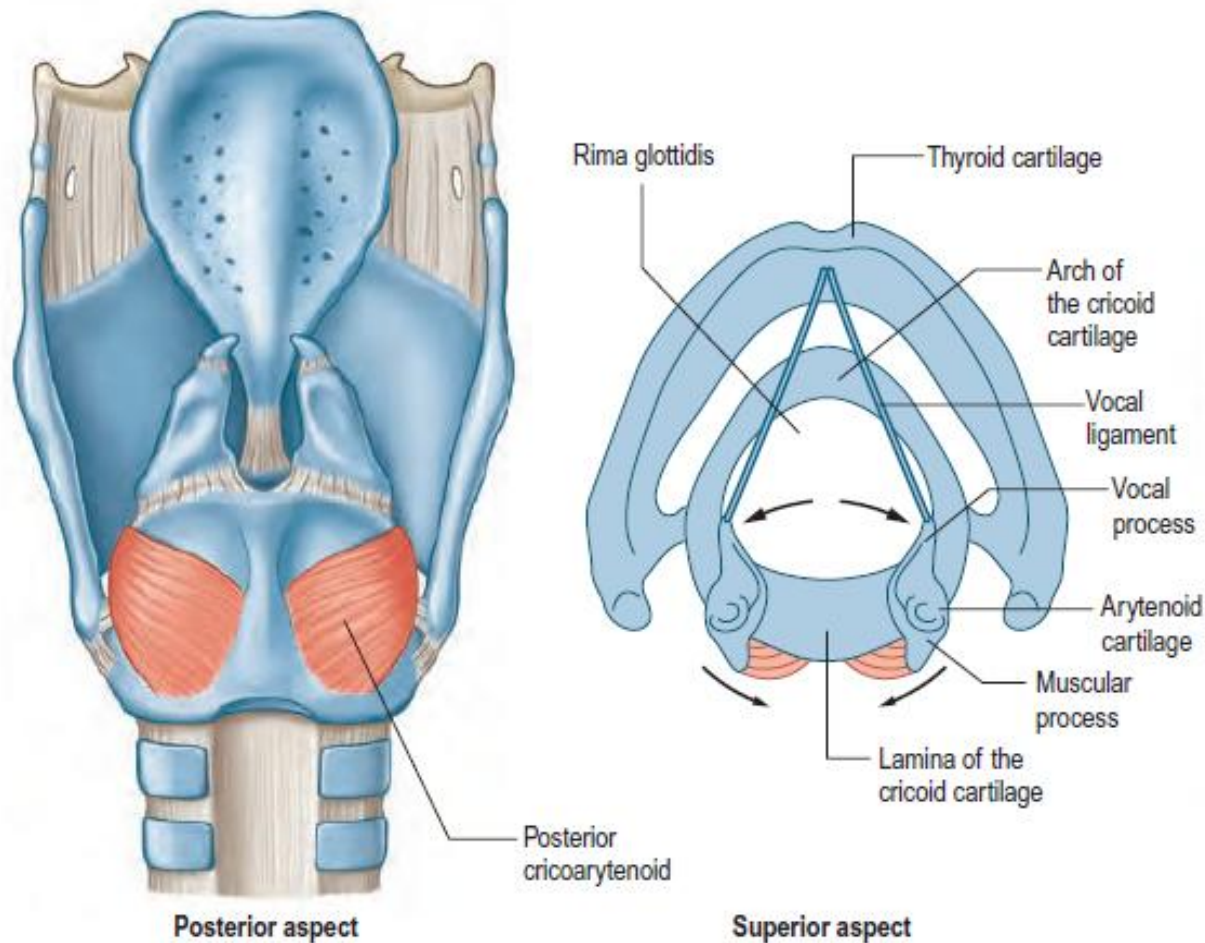
- Thyroarytenoids
- lateral to laryngeal ventricles and saccules
 - close rima glottidis
- Vocalis
- lateral and parallel to each vocal ligament
 - shorten (relax) the vocal ligaments



(B) Superior view
Thyro-arytenoid

Posterior cricoarytenoid muscles

- **Abduct** vocal ligaments and open rima glottidis
 - produce unvoiced sounds



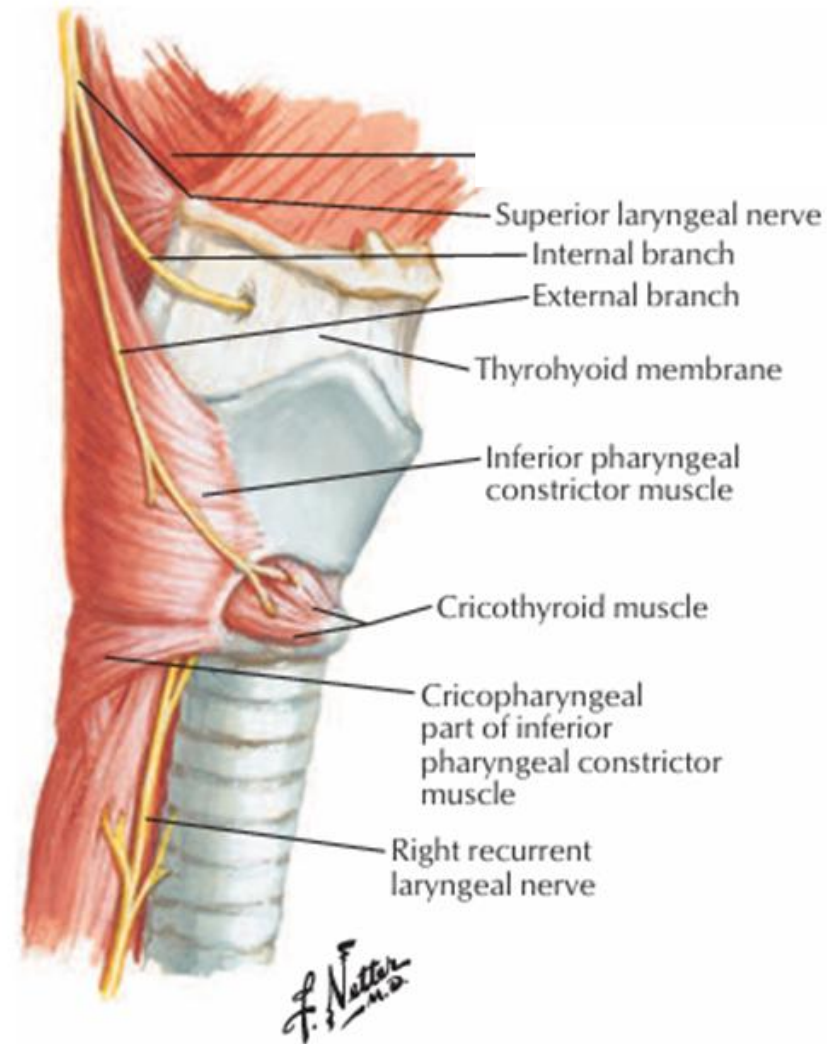
Laryngeal Innervation

Sensory

- Above vocal cords:
internal laryngeal nerve
- Below vocal cords:
recurrent laryngeal nerve

Motor

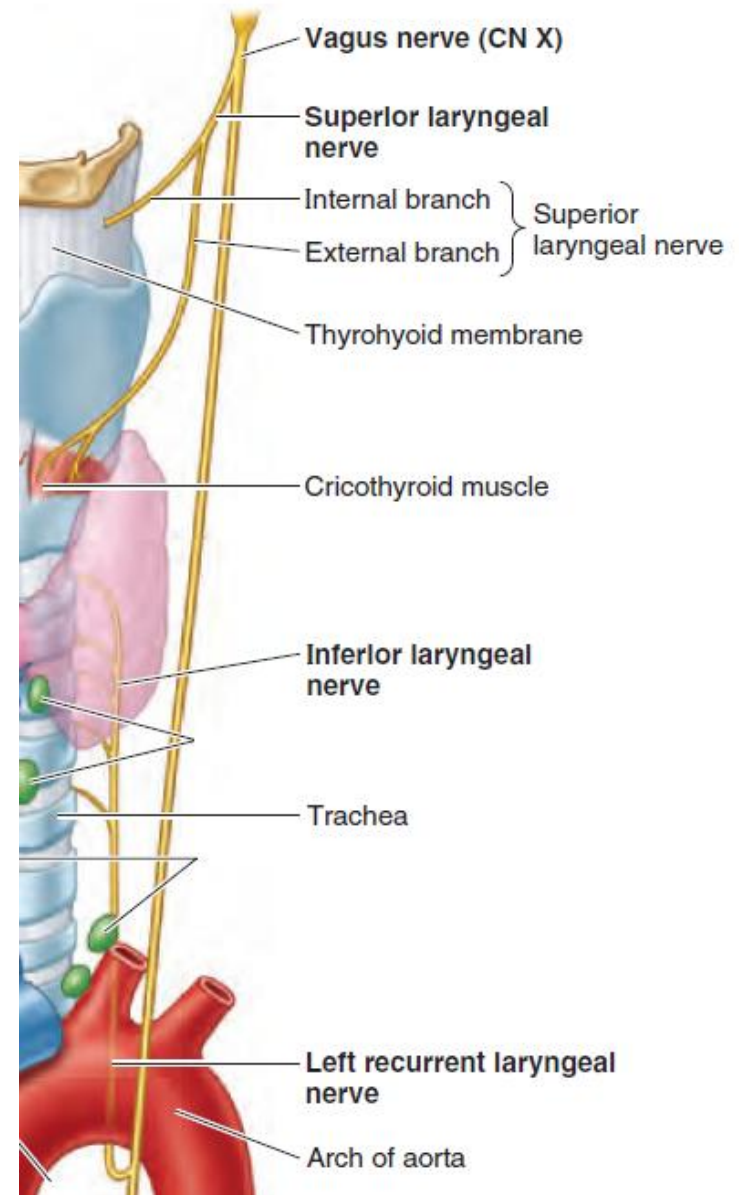
- Most intrinsic muscles of the larynx:
recurrent laryngeal nerve
- Exception: **cricothyroid**
 - supplied by
external laryngeal nerve



Clinical Neuropathy

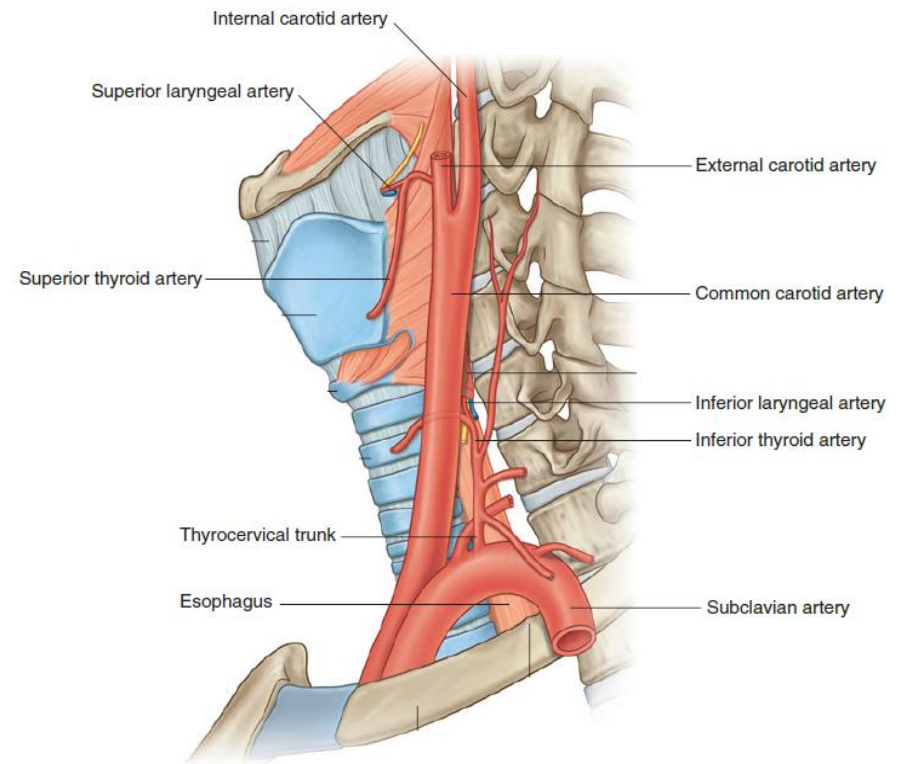
Patient presentation:

- Lesion of **internal laryngeal**
 - anesthesia of superior larynx
- Lesion of **external laryngeal**
 - weak voice that tires easily
- Lesion of **recurrent laryngeal**
 - unilateral: dysphonia
 - bilateral: aphonia



Larynx - Arterial supply

- Superior laryngeal artery
 - branch of **superior thyroid artery**
 - from *external carotid*
- Inferior laryngeal artery
 - branch of **inferior thyroid artery**
 - from *thyrocervical trunk*



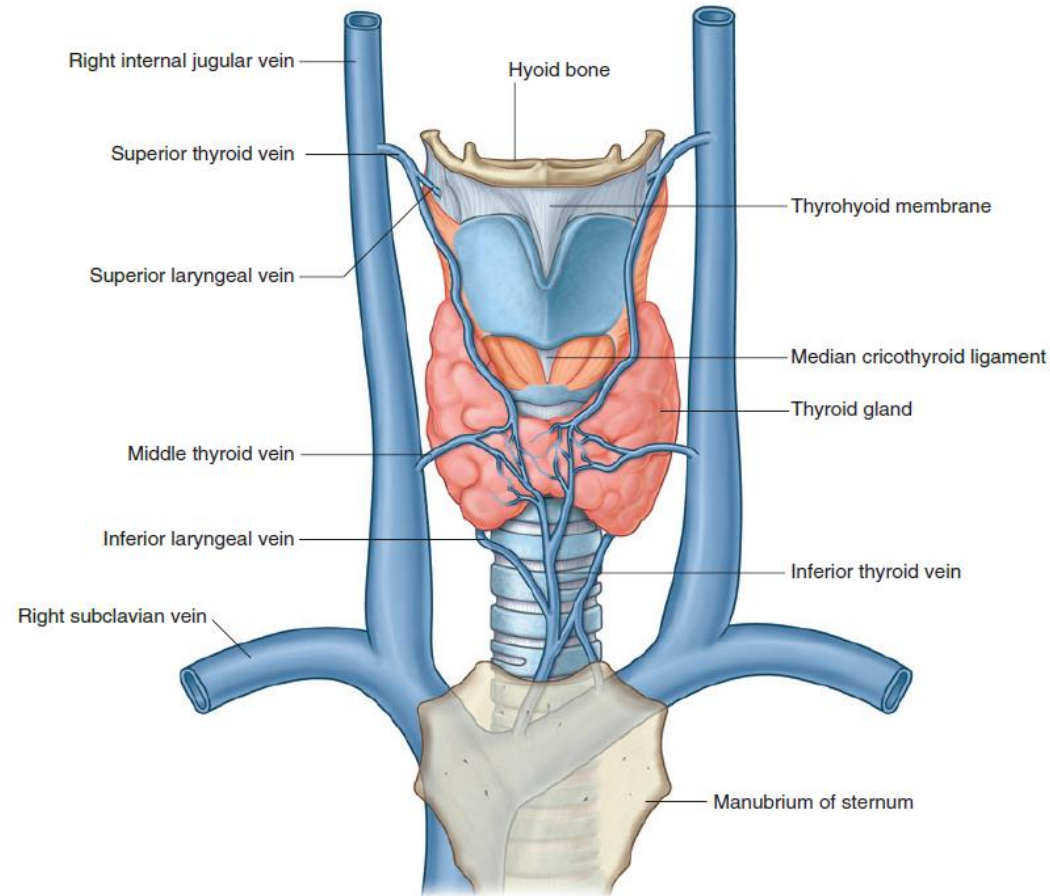
Larynx - Venous drainage

- Superior laryngeal veins

- drain into **superior thyroid vein**
 - drain into *internal jugular*

- Inferior laryngeal veins

- drain into **inferior thyroid vein**
 - drain into *left brachiocephalic*



Larynx - Lymphatic drainage

- Above vocal cords:
 - to deep cervical nodes along common carotid bifurcation
- Below vocal cords:
 - to deep cervical nodes associated with inferior thyroid

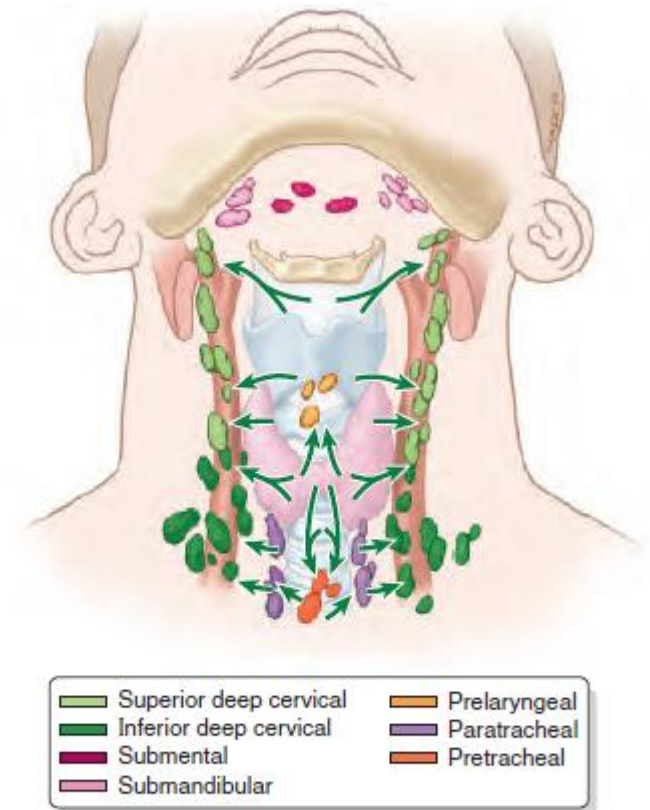
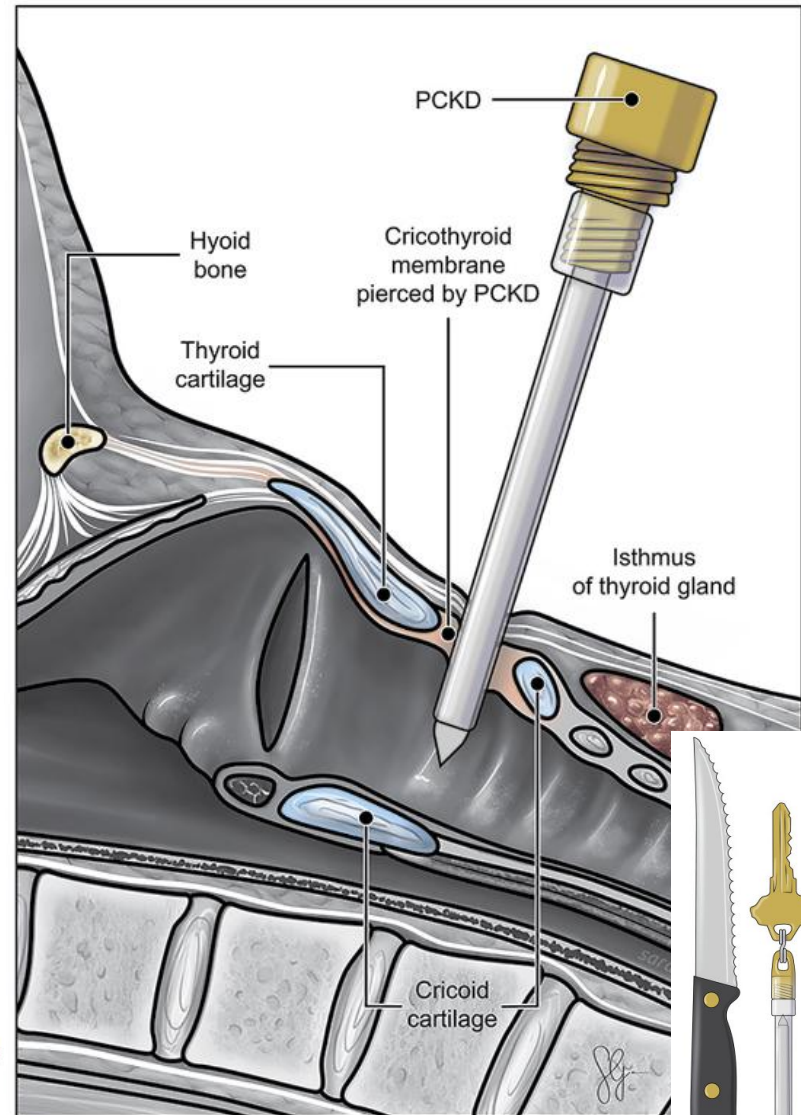
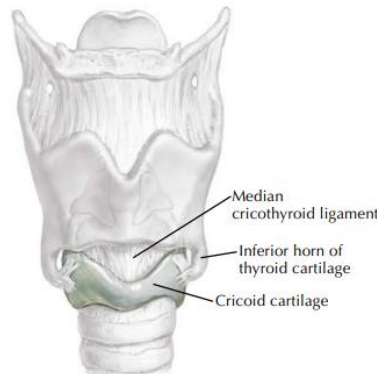


FIGURE 8.29. Lymphatic drainage of thyroid gland, larynx, and trachea. The arrows indicate the direction of lymph flow.

Airway Access - Cricothyroidotomy

- Emergency procedure for a person with a blocked airway above the vocal folds level
- Perforation of **median cricothyroid ligament** to establish an airway
 - Few structures to damage
 - small vessels
 - possible pyramidal lobe of thyroid gland

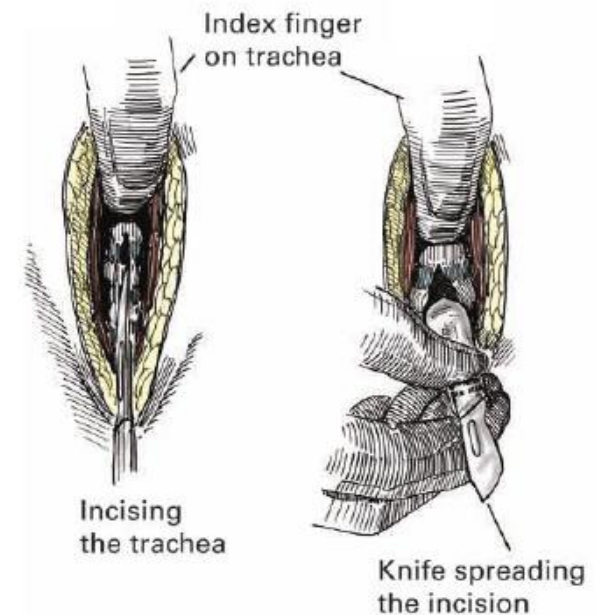
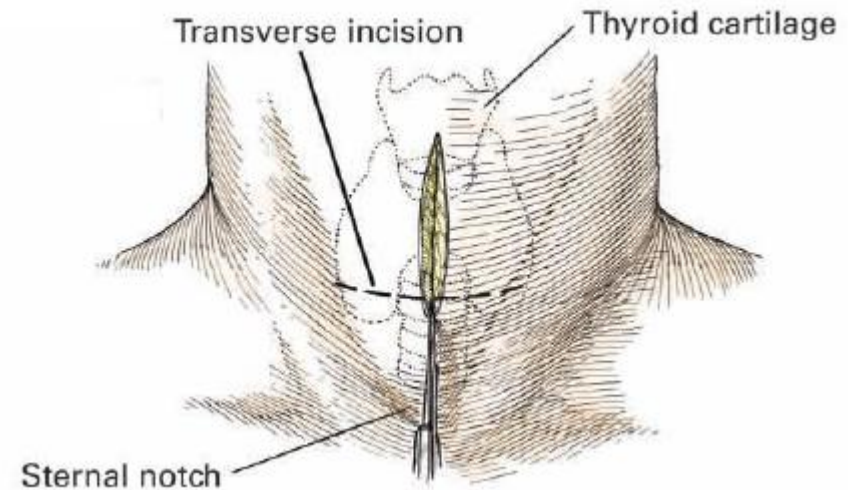


DOI:10.1016/j.tria.2020.100070



Airway Access - Tracheotomy

- Common in hospital settings
 - Can be done with full or local anesthetic
- Ideal for temporary or permanent intubation
- May involve division of **thyroid isthmus** and **tracheal cartilage**
 - caution with *inferior thyroid vein*



fin

